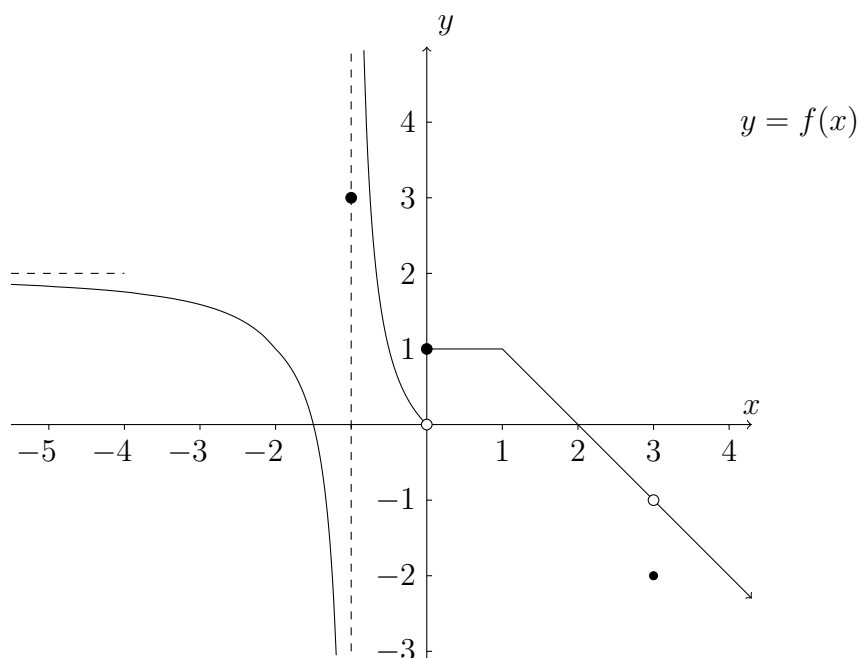


1. (4 points) Use the graph of the function  $y = f(x)$  below to answer the questions below. No work needs to be shown.



Compute the following. Write DNE or  $-\infty$  or  $\infty$  as it applies.

- |                                               |                                                            |
|-----------------------------------------------|------------------------------------------------------------|
| (a) $\lim_{x \rightarrow -\infty} f(x)$ _____ | (e) $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$ _____ |
| (b) $\lim_{x \rightarrow -1^+} f(x)$ _____    | (f) $\lim_{x \rightarrow 3} f(x)$ _____                    |
| (c) $f(-1)$ _____                             | (g) $\lim_{x \rightarrow \infty} f(x)$ _____               |
| (d) $\lim_{x \rightarrow 0} f(x)$ _____       | (h) $\lim_{x \rightarrow -1^-} \frac{1}{f(x)}$ _____       |

2. (2 points) For the graph of question 1, determine:

- The values of  $x$  where the function  $f(x)$  is not differentiable.
- The intervals where  $f'(x) < 0$ .

3. (18 points) Evaluate the following limits. Use the algebraic methods whenever possible. Identify the limits that do not exist and use  $+\infty$  or  $-\infty$  where appropriate. **Justify your answer and show your work.**

- |                                                                       |                                                                                                 |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| a) $\lim_{x \rightarrow 7} \frac{2x^2 - 15x + 7}{x - 7}$              | b) $\lim_{x \rightarrow -5} \frac{x^2 + 1}{x + 5}$                                              |
| c) $\lim_{x \rightarrow 3} \frac{\frac{4}{x+3} - \frac{2}{3}}{x - 3}$ | d) $\lim_{x \rightarrow -\infty} \frac{(4x^3 + 7)(5x^6 + 2x^3 + 3x - 1)}{(5x + 3)(6x^2 - x^6)}$ |

e)  $\lim_{x \rightarrow 0} \frac{5 - \sqrt{x + 25}}{3x}$

f)  $\lim_{x \rightarrow 1^-} \frac{|x - 1|}{x^2 + 6x - 7}$

4. (5 points) Determine the value(s) of  $x$  at which the given function is discontinuous. Justify your answer using the definition of continuity.

$$f(x) = \begin{cases} \frac{3}{x^2 - 4} & \text{if } x < 0 \\ 1 - \frac{x}{5} & \text{if } 0 \leq x \leq 4 \\ \frac{2x - 10}{(x - 5)(x + 6)} & \text{if } x > 4 \end{cases}$$

5. (3 points) Using the definition of continuity, find the value(s) of  $k$  such that the given function is continuous everywhere.

$$f(x) = \begin{cases} k^2x + 2x & \text{if } x \leq 1 \\ x + k + 1 & \text{if } x > 1 \end{cases}$$

6. (4 points) (a) Use the limit definition of the derivative to find  $f'(x)$  given that  $f(x) = 5x - 3x^2$ .  
(b) Check your answer to part (a) using appropriate differentiation rules.

7. (20 points) Find  $y'$  for each of the following. **Do not simplify your answers.**

a)  $y = \frac{7}{\sqrt{x^5}} - 2e^x + 3x^e + \pi^2$

b)  $y = \frac{\csc(3x)}{\sqrt{x} + \log_5(x^2 + 1)}$

c)  $y = (3^{x^2+5} + 1) \cot(7x)$

d)  $y = \sqrt{\ln[\sec(x)]}$

e)  $e^{xy} - 4y = y \ln(x) + e^2$

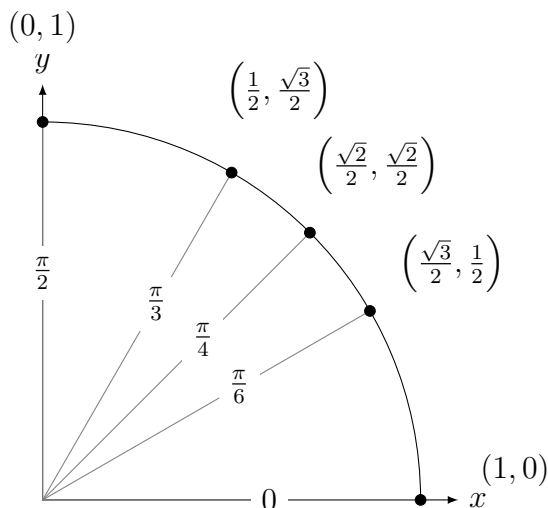
f)  $y = 5x^{6x+1}$

8. (4 points) **Use the properties of logarithms to rewrite the function** and then find its derivative.

$$y = \ln \left( \frac{\sin^3(5x) \sqrt[5]{3x^2 - 1}}{(e^{7x+1} + \cos x)^4} \right)$$

9. (3 points) Find the 211<sup>th</sup> derivative of  $y = \cos(1 - x) + e^{-2x} + x^{210}$ .

10. (3 points) Find an equation for the tangent line to the graph of  $y = \tan^2 x$  at  $x = \frac{5\pi}{4}$ . Use exact values.



11. (4 points) Time<sup>2</sup> manufactures smartwatches. The demand is  $p = 200 - 4x$ , where  $x$  is the number sold per month and  $p$  is the price per watch in dollars. The cost function is  $C(x) = 0.5x^3 - 10x^2 + 150x + 500$ .
- (a) Find the revenue function  $R(x)$ .
  - (b) Find the profit function  $P(x)$ .
  - (c) Find the marginal profit function.
  - (d) Determine the marginal profit when  $x = 10$  and interpret your result.

12. (4 points) Find the absolute extrema of  $f(x) = \frac{x^2 + 25}{4x}$  on the interval  $[2, 6]$ .

13. (11 points) Given

$$f(x) = \frac{1}{x^2 - x}, \quad f'(x) = \frac{1 - 2x}{(x^2 - x)^2} \quad \text{and} \quad f''(x) = \frac{2(3x^2 - 3x + 1)}{(x^2 - x)^3}, \text{ find:}$$

- (a) The domain.
  - (b) The  $x$  and  $y$  intercepts.
  - (c) The vertical and horizontal asymptotes.
  - (d) The intervals of which  $f(x)$  is increasing or decreasing.
  - (e) The relative extrema.
  - (f) The intervals of upward and downward concavity.
  - (g) The points of inflection.
  - (h) On the next page, sketch the graph of  $f(x)$ . Label all intercepts, extrema and points of inflection. Draw your sketch.
14. (5 points) Robins Kitchen Company sells 32 air fryers per month at a price of \$66 each. If they drop the price by \$3 per unit, they will sell 4 more units per month. What should be the unit price of an air fryer in order to maximize their revenue. (Be sure to use a test to confirm that this is a maximum.)
15. (5 points) A storage box with square base and no top is to have a volume of  $40 \text{ m}^3$ . Material for the base costs \$5 per square meter. Material for the sides costs \$4 per square meter. Find the dimensions of the least expensive container that can be constructed. What would be its construction cost? (Be sure to use a test to confirm that this is a minimum.)
16. (5 points) Suppose that demand for tickets in the 100-level at the Bell Centre can be given by the function  $x = \frac{1}{3}(3000 - 80\sqrt{p})$ .
- (a) Find the elasticity of demand function.
  - (b) Find the elasticity of demand at  $p = \$900$ . Is the demand elastic or inelastic?
  - (c) If the price of \$900 were to be decreased by 6%, how would demand be affected?
  - (d) What price would maximize revenue?

**Answers**

1. (a) 2 (b)  $\infty$  (c) 3 (d) DNE (e)  $-1$  (f)  $-1$  (g)  $-\infty$  (h) 0

2. (a)  $x = -1, 0, 1, 3$  (b)  $(-\infty, -1), (-1, 0), (1, 3), (3, \infty)$

3. (a) 13 (b) DNE (c)  $-1/9$  (d)  $-\infty$  (e)  $-1/30$  (f)  $-1/8$

4.  $x = -2$  since  $f(-2)$  is undefined,  $x = 0$  since  $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$  and  $x = 5$  since  $f(5)$  is undefined.

5.  $k = 0, k = 1$  6. (a)  $f'(x) = 5 - 6x$

7. (a)  $y' = \frac{-35}{2x^{7/2}} - 2e^x + 3ex^{e-1}$  (b)  $y' = \frac{-3 \csc(3x) \cot(3x)(\sqrt{x} + \log_5(x^2 + 1)) - \csc(3x) \left( \frac{1}{2\sqrt{x}} + \frac{2x}{(x^2+1)\ln 5} \right)}{(\sqrt{x} + \log_5(x^2 + 1))^2}$

(c)  $y' = 3^{x^2+5}(2x)(\ln 3) \cot(7x) - 7(3^{x^2+5} + 1) \csc^2(7x)$  (d)  $y' = \frac{\tan(x)}{2\sqrt{\ln[\sec(x)]}}$

(e)  $y' = \frac{y/x - ye^{xy}}{xe^{xy} - \ln x - 4}$  (f)  $y' = 5x^{6x+1} \left( 6 \ln x + \frac{6x+1}{x} \right)$

8.  $y' = 15 \cot(5x) + \frac{6x}{5(3x^2 - 1)} - \frac{4(7e^{7x+1} - \sin x)}{e^{7x+1} + \cos x}$  9.  $y = -\sin(1 - x) - 2^{211}e^{-2x}$  10.  $y = 4x - 5\pi + 1$

11. (a)  $R(x) = 200x - 4x^2$  (b)  $P(x) = -0.5x^3 + 6x^2 + 50x - 500$  (c)  $P'(x) = -1.5x^2 + 12x + 50$

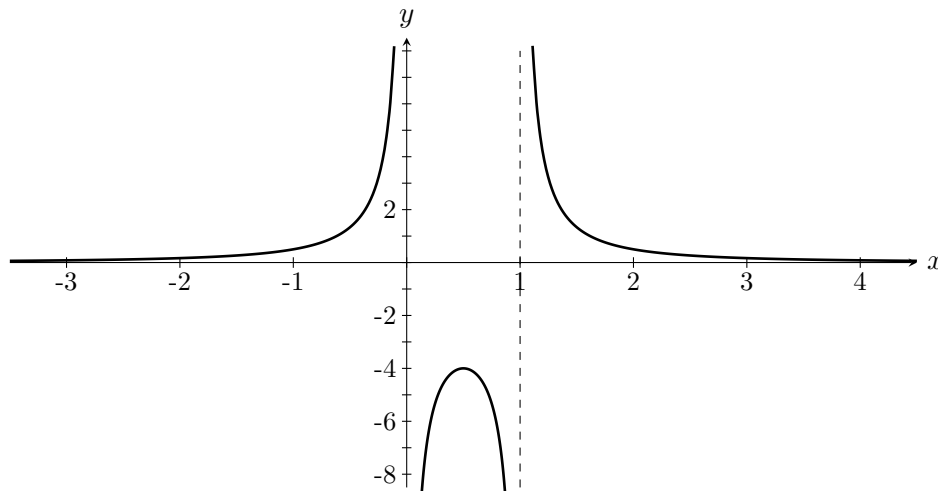
(d)  $P'(10) = 20$ ; The profit obtained from selling the 11th smartwatch is approximately \$20.

12. Absolute minimum at  $(5, 2.5)$ , absolute maximum at  $(2, 3.625)$

13. (a)  $\mathbb{R} \setminus \{0, 1\}$  (b) None (c)  $x = 0, x = 1, y = 0$  (d) Increasing:  $(-\infty, 0) \cup (0, \frac{1}{2})$  Decreasing:  $(\frac{1}{2}, 1) \cup (1, \infty)$

(e) Relative max:  $(\frac{1}{2}, -4)$  (f) Upward:  $(-\infty, 0), (1, \infty)$  Downward:  $(0, 1)$  (g) None

(h)



14. \$45 15. (a)  $4 \text{ m} \times 4 \text{ m} \times 2.5 \text{ m}$  (b) \$240

16. (a)  $E(p) = \frac{40\sqrt{p}}{3000 - 80\sqrt{p}}$  (b)  $E(900) = 2$ , Elastic (c) Demand would increase by 12%. (d) \$625